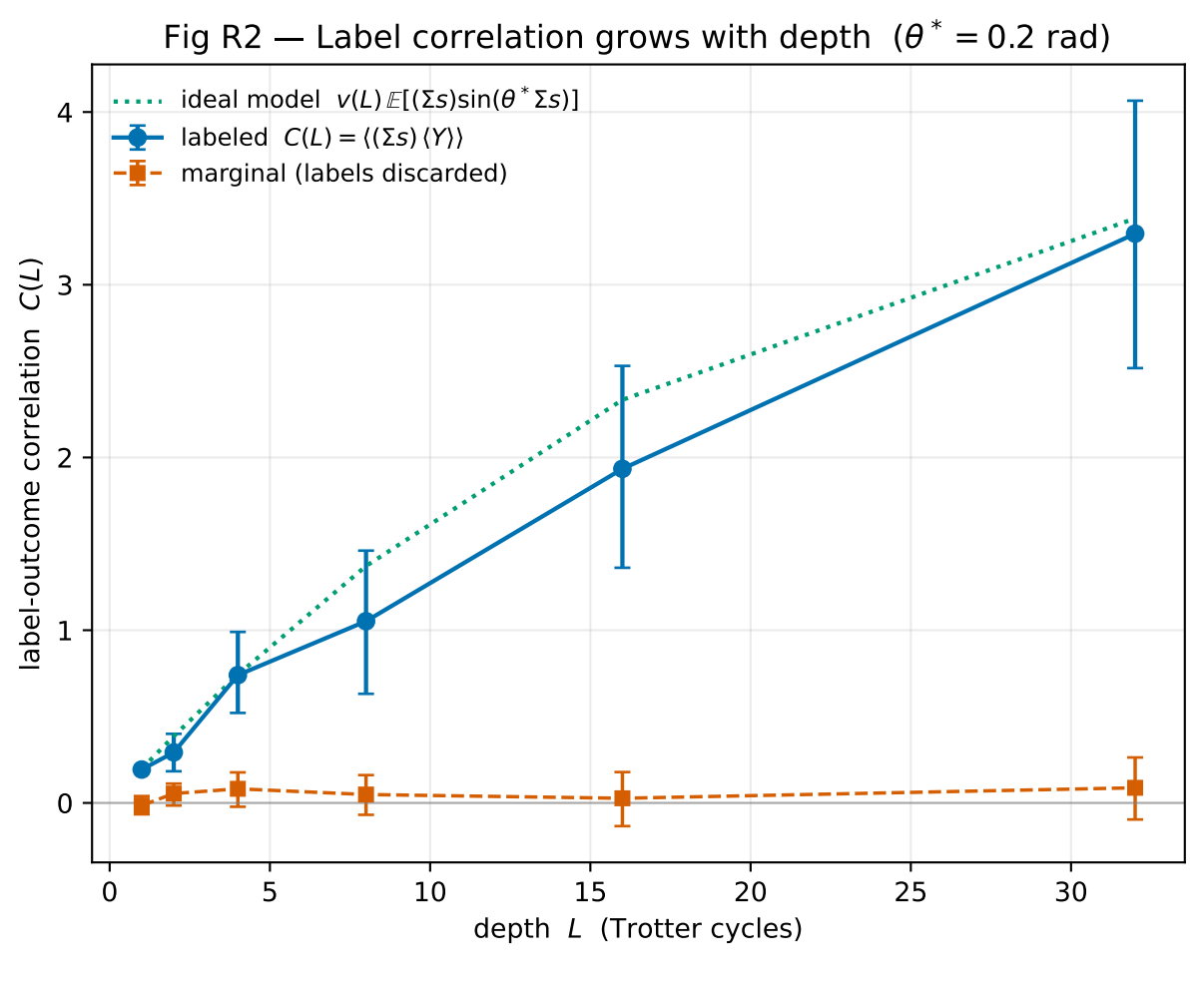
Fig R2 — Label correlation grows with depth ($\theta^* = 0.2$ rad)
ideal model $v(L)\,\mathbb{E}[(\Sigma s)\sin(\theta^*\Sigma s)]$
labeled $C(L) = \langle(\Sigma s)\langle Y\rangle\rangle$
marginal (labels discarded)
label-outcome correlation $C(L)$
depth L (Trotter cycles)
0
1
2
3
4
0
5
10
15
20
25
30
backend=ibm_marrakesh, qubit=2, date=2026-07-21, shots=3000/circuit, 50 twirl instances/point